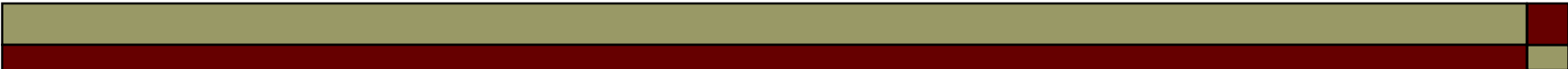


Software Construction and Development SE-312

LECTURE # 01

Introduction to Software Construction

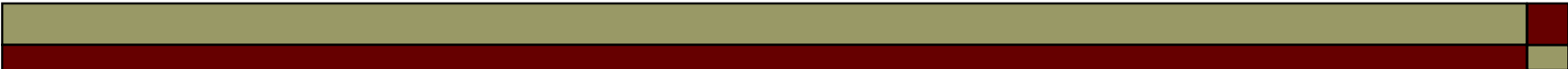
Course Instructor: Engr. Ruqqiya Asad



Objectives

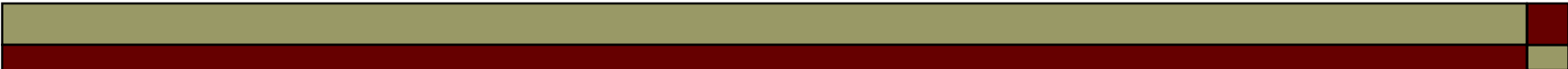
Contents

- ❑ • What is Software Construction
- ❑ • Importance of Software Construction
- ❑ • SWEBOK Guide (2004 Edition)
 - Fundamentals of Software Construction
 - Software Construction Management
- ❑ • Key points of Software Construction



Why This Course?

- Ever used an app that crashes?
 - App that works today but fails after update?
 - Software failure is **not only coding issue**
 - It is a **construction & development problem**
- This course teaches **how to build quality software**



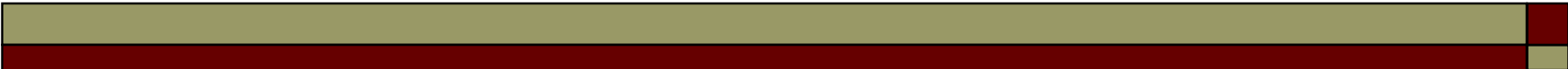
What is Software Construction & Development?

- Software construction is more **than coding**
- Focuses on:
 - Planning
 - Writing clean code
 - Error handling
 - Testing
 - Maintenance

Key Idea:

Good code works today

Good construction works forever



Software Construction

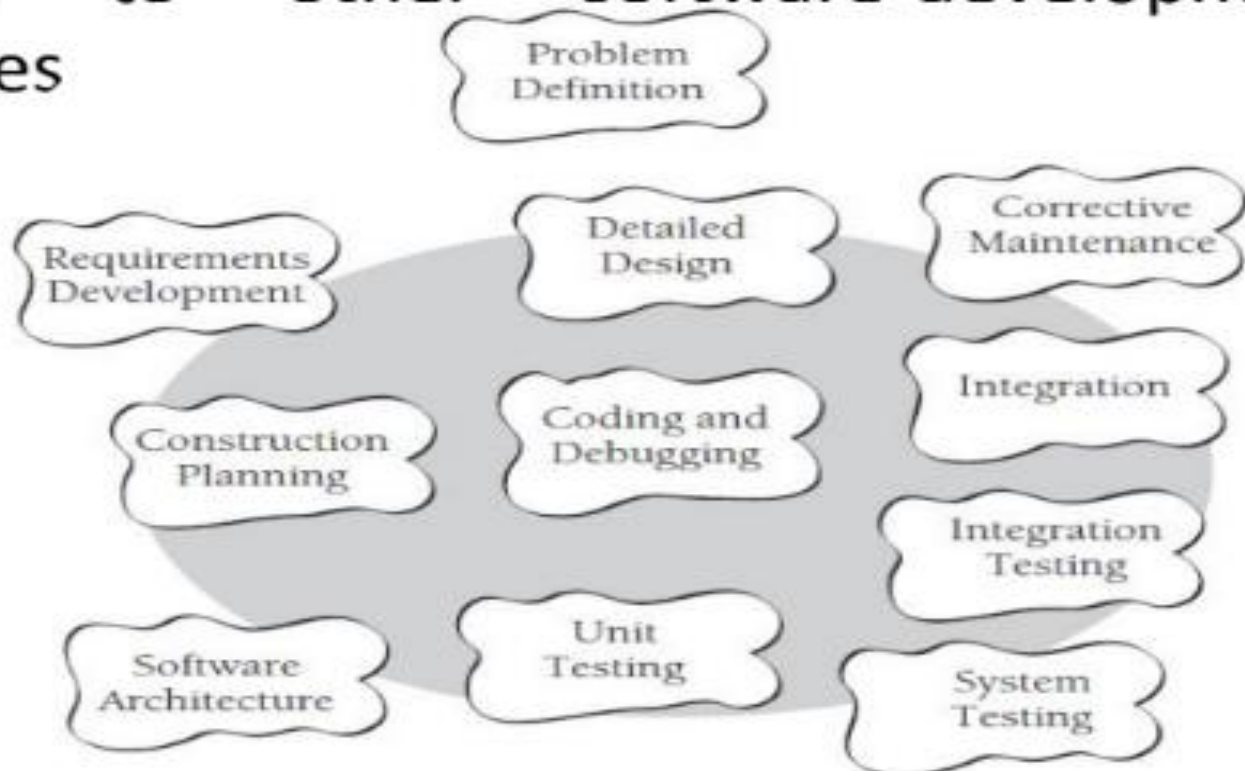
- Construction is the work “**construction workers**” do when they build a house, a school, or a skyscraper.
- In common usage, construction refers to the process of building.
- The construction process might include some aspects of planning, designing, and checking the work, but mostly construction refers to the hands-on part of creating something.

Software Construction

- ❑ Software construction is a software engineering discipline.
- ❑ It is detailed creation of working, meaningful software through a combination of coding, verification, unit testing, integration testing, and debugging.

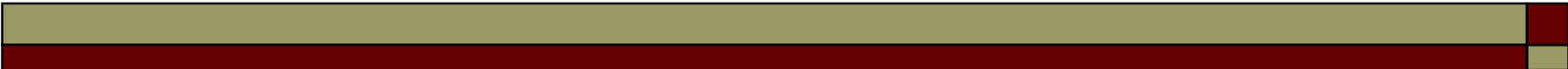
Software Construction

- Following figure illustrates construction's place related to other software-development activities



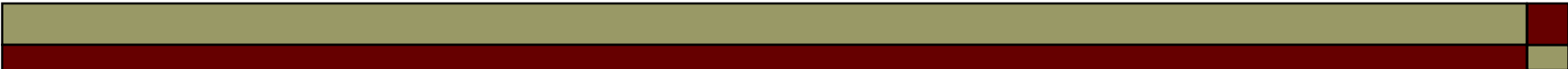
Software Construction

- Some of the specific tasks involved in construction are:
 - ❑ Verifying that the groundwork has been laid so that construction can proceed successfully
 - ❑ Determining how your code will be tested
 - ❑ Designing and writing classes and routines
 - ❑ Creating and naming variables and named constants
 - ❑ Selecting control structures and organizing blocks of statements
 - ❑ Unit testing, integration testing, and debugging your own code
 - ❑ Reviewing other team members' low-level designs and code and having them review yours
 - ❑ Polishing code by carefully formatting and commenting it
 - ❑ Integrating software components that were created separately
 - ❑ Tuning code to make it faster and use fewer resources



Software Construction

- Important non-construction activities include
 - ❑ Management
 - ❑ requirements development
 - ❑ software architecture
 - ❑ user-interface design
 - ❑ system testing
 - ❑ maintenance.



Software Construction

- Construction is also sometimes known as “coding” or “programming.”

Do you think both these words correctly portray the term construction?



Importance of Software Construction

- Software Construction is an important part of software development because
 - ❑ Construction is a large part of software development
 - ❑ Construction is the central activity in software development
 - ❑ With a focus on construction, the individual programmer's productivity can improve enormously
 - ❑ Construction's product, the source code, is often the only accurate description of the software
 - ❑ Construction is the only activity that's guaranteed to be done

